ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 1ST YEAR

ELECTRICAL CIRCUITS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 1st year, semester 2 final examination: basic quantities and resistance, series-parallel circuits, voltage division, Kirchhoff's laws, nodal analysis, Thevenin and Norton equivalents, maximum power transfer, operational amplifiers, capacitors and inductors, single-phase AC circuits.

Written by Academic Central for practice, following first-year Electrical Circuits syllabuses at South African universities (DC and AC circuit analysis for engineering students). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Show ALL your working, including formulae and substitutions. Give answers correct to TWO decimal places, with units.
3. Draw circuit diagrams neatly in pencil, using standard symbols.
4. Non-programmable scientific calculators may be used.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: BASIC QUANTITIES AND RESISTANCE

1.1 Define electric current, potential difference and resistance, and give the SI unit of each. (3)

1.2 A copper cable is 100 m long and has a cross-sectional area of 1.5 mm². The resistivity of copper is 1.72×10^{-8} Ω·m. Calculate the resistance of the cable. (3)

1.3 A 4 kΩ and an 8 kΩ resistor are connected in series across a 12 V supply. Use the voltage divider rule to calculate the voltage across the 8 kΩ resistor. (2)

1.4 A 12 V battery delivers a steady current of 2.5 A for 3 hours. Calculate:

1.4.1 the charge that flows (2)

1.4.2 the energy delivered, in joules and in kWh (2)

1.5 A resistor has the colour bands brown, black, red, gold.

1.5.1 Write down its resistance and tolerance. (1)

1.5.2 It is connected across 12 V. Show whether a 0.25 W resistor is suitable. (2)

[15]

QUESTION 2: SERIES-PARALLEL CIRCUITS

Answer the questions using the circuit in FIGURE 1. The supply has no internal resistance.

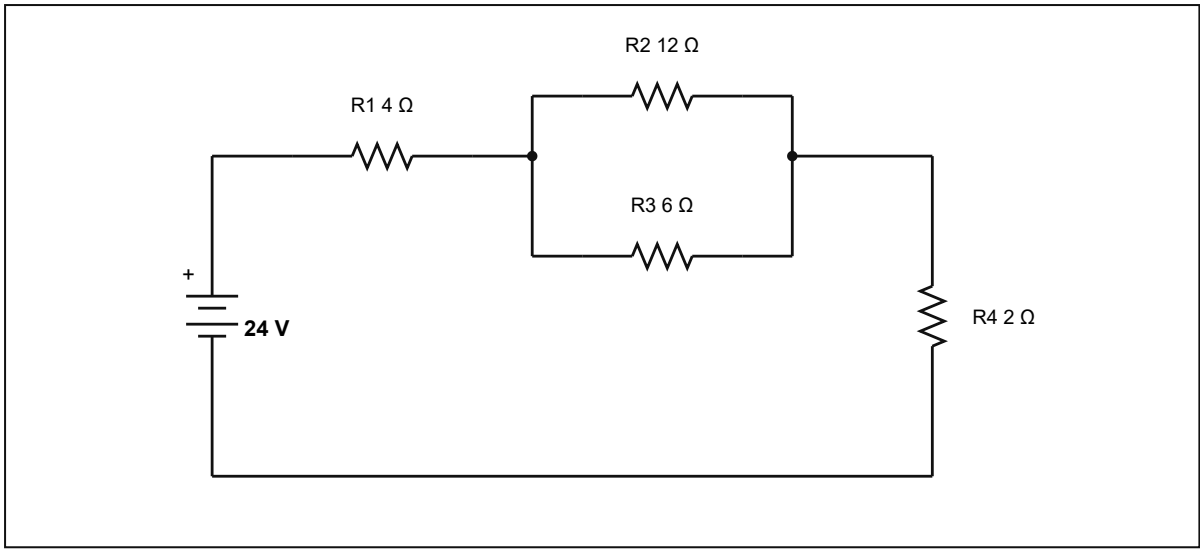


FIGURE 1: Series-parallel circuit

2.1 Calculate the total resistance of the circuit. (4)

2.2 Calculate the current drawn from the supply. (2)

2.3 Calculate the voltage across R1, across the parallel group R2 ∥ R3, and across R4. (4)

2.4 Calculate the current through R2 and through R3. (3)

2.5 Calculate the total power supplied and the power used by R3. (4)

2.6 Show that Kirchhoff's voltage law holds around the loop that passes through R1, R2 and R4. (3)

[20]

QUESTION 3: NETWORK THEOREMS AND OPERATIONAL AMPLIFIERS

3.1 An inverting amplifier is built with an ideal op-amp, an input resistor Rin = 10 kΩ and a feedback resistor Rf = 100 kΩ.

3.1.1 Calculate the voltage gain and the output voltage when the input is 0.5 V DC. (2)

3.1.2 Give TWO properties of an ideal op-amp. (2)

Answer QUESTIONS 3.2 to 3.4 using the circuit in FIGURE 2. The sources have no internal resistance.

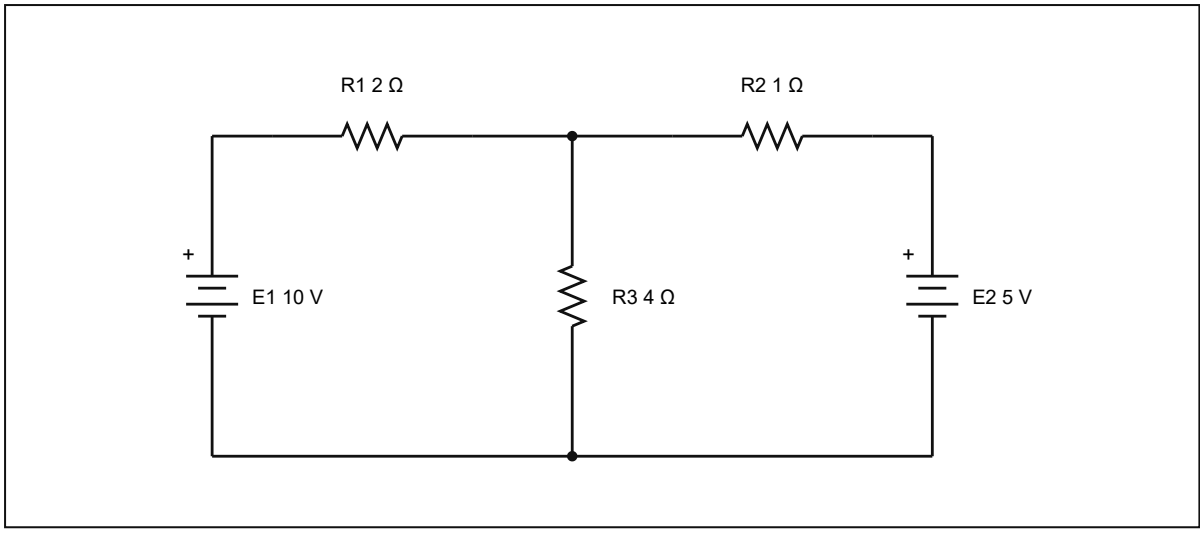


FIGURE 2: Network with two sources

3.2 Use Kirchhoff's laws (or nodal analysis):

3.2.1 Calculate the voltage across R3 and the current through R3. (6)

3.2.2 Calculate the current in each source and state its direction. (3)

3.3 Use Thevenin's theorem, with R3 as the load:

3.3.1 Calculate the Thevenin voltage Vth. (3)

3.3.2 Calculate the Thevenin resistance Rth. (2)

3.3.3 Calculate the current through R3 and compare it with your answer in 3.2.1. (2)

3.3.4 Convert the Thevenin equivalent into a Norton equivalent: give the Norton current and the Norton resistance. (2)

3.4 What value of load resistance in place of R3 would receive the maximum power, and what is that maximum power? (3)

[25]

QUESTION 4: CAPACITORS AND INDUCTORS

4.1 A 470 µF capacitor is charged to 12 V. Calculate the charge stored and the energy stored. (4)

4.2 Capacitors of 10 µF, 20 µF and 30 µF are available. Calculate the total capacitance when they are connected:

4.2.1 in series (2)

4.2.2 in parallel (1)

4.3 A 100 µF capacitor is charged through a 10 kΩ resistor from a 12 V DC supply. The capacitor starts uncharged.

4.3.1 Calculate the time constant. (1)

4.3.2 Calculate the capacitor voltage 2 s after the switch is closed. (3)

4.3.3 After about how long can the capacitor be taken as fully charged? (1)

4.3.4 Sketch the capacitor voltage and the charging current against time. (2)

4.4 The current in a 0.5 H inductor rises steadily from 0 A to 2 A in 10 ms. Calculate the induced emf and the energy stored at 2 A. (4)

4.5 Explain why a diode is connected across the coil of a relay that is switched by a transistor. (2)

[20]

QUESTION 5: SINGLE-PHASE AC CIRCUITS

5.1 An alternating voltage is given by v = 325 sin(314.16t) V. Write down or calculate the peak voltage, the RMS voltage, the frequency and the period. (4)

5.2 A series circuit has R = 40 Ω, L = 0.1 H and C = 50 μF, connected to a 230 V, 50 Hz supply.

5.2.1 Calculate the inductive reactance and the capacitive reactance. (2)

5.2.2 Calculate the impedance of the circuit. (3)

5.2.3 Calculate the current. (1)

5.2.4 Calculate the phase angle and the power factor, and state whether the current leads or lags the voltage. (3)

5.2.5 Calculate the true (active) power. (2)

5.2.6 Calculate the resonant frequency of the circuit. (3)

5.3 Draw the phasor diagram for the circuit in QUESTION 5.2, using the current as the reference. (2)

[20]

TOTAL: 100

END OF QUESTION PAPER